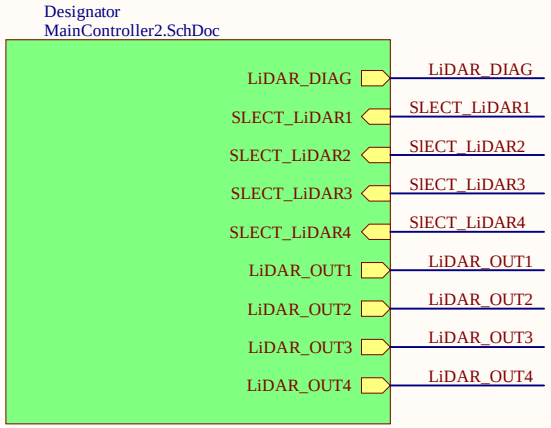
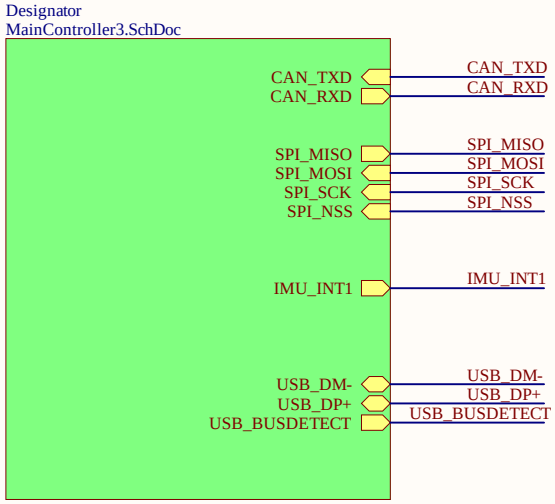
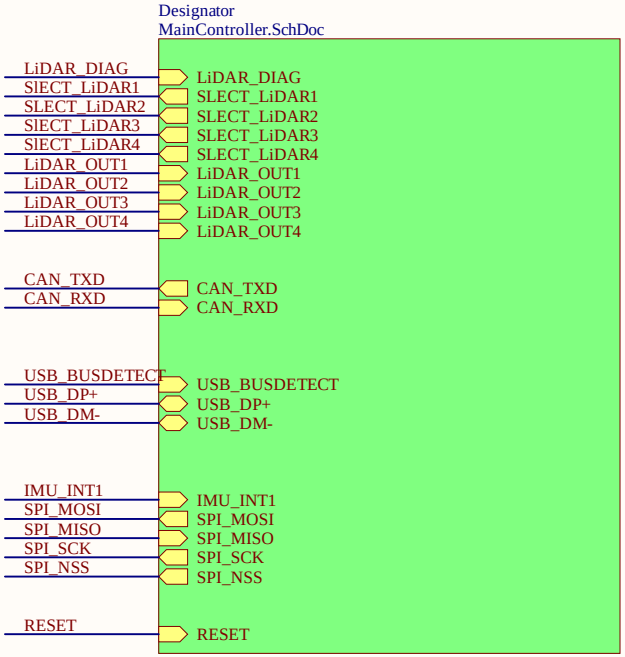
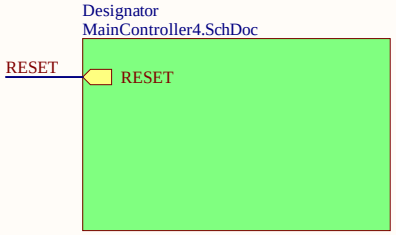
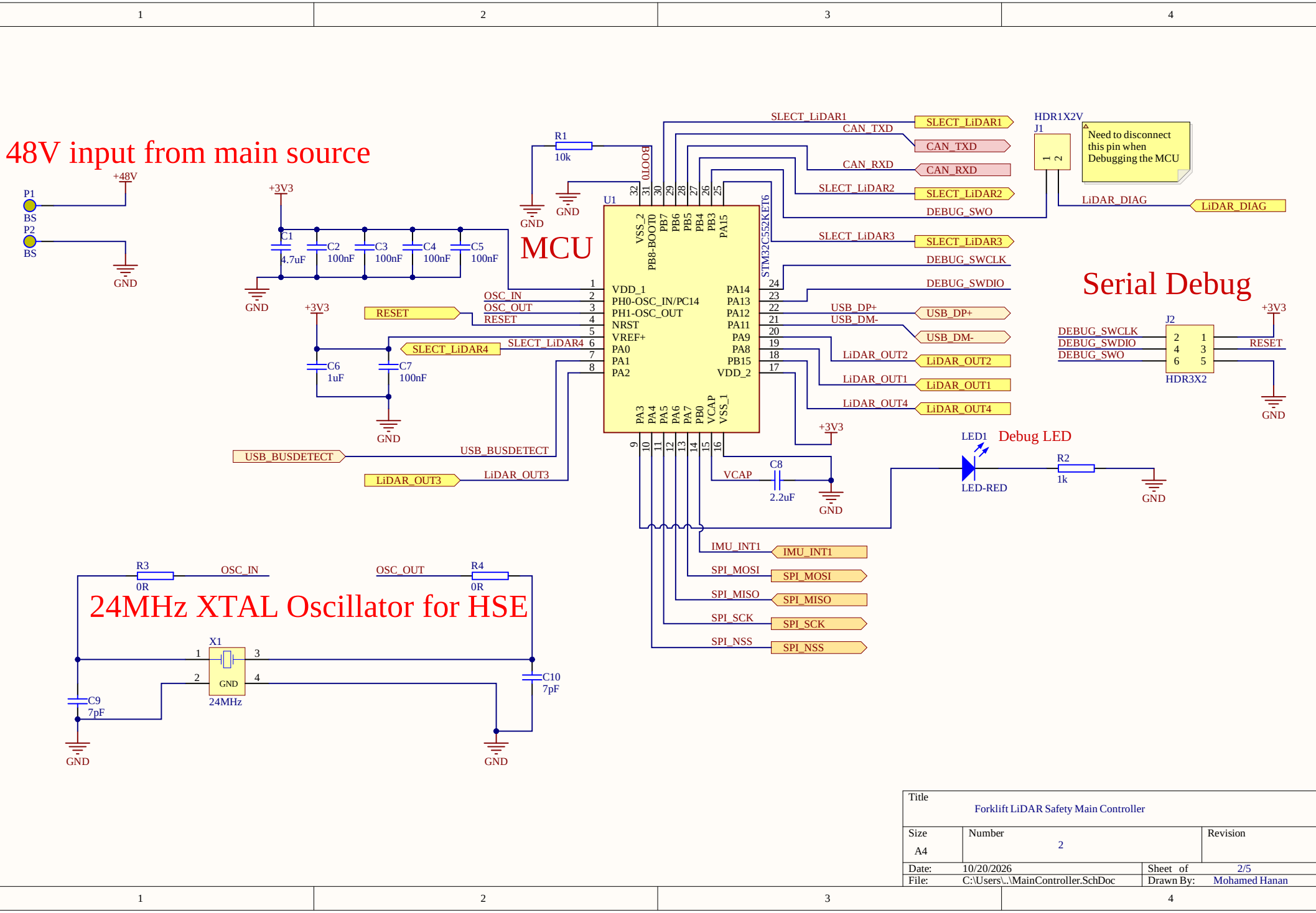






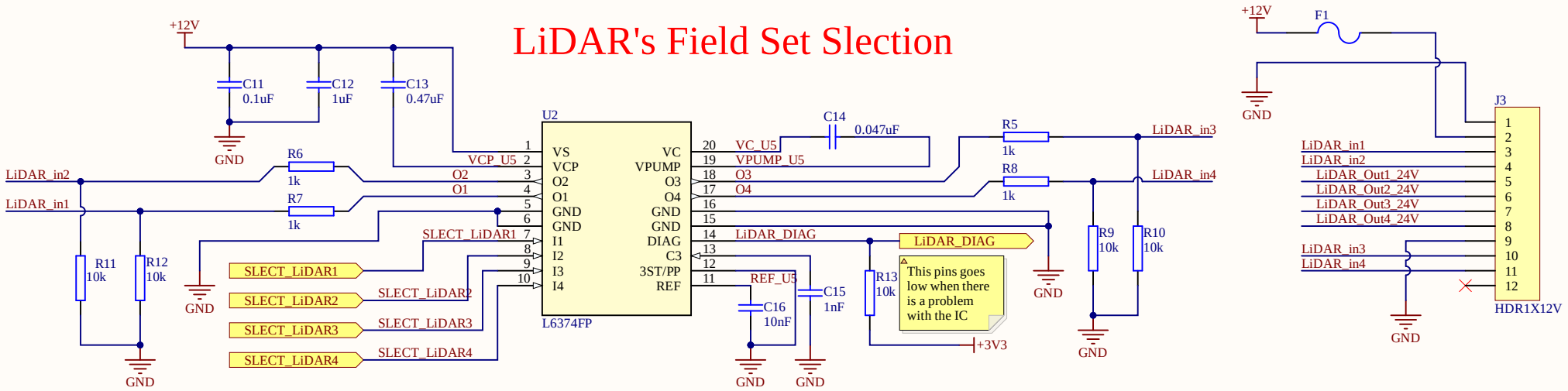


Title: Forklift LiDAR Safety Main Controller		
Size: A4	Number: 1	Revision:
Date: 10/20/2026		Sheet of 1/5
File: C:\Users\..\ConnectionMain.SchDoc		Drawn By: Mohamed Hanan

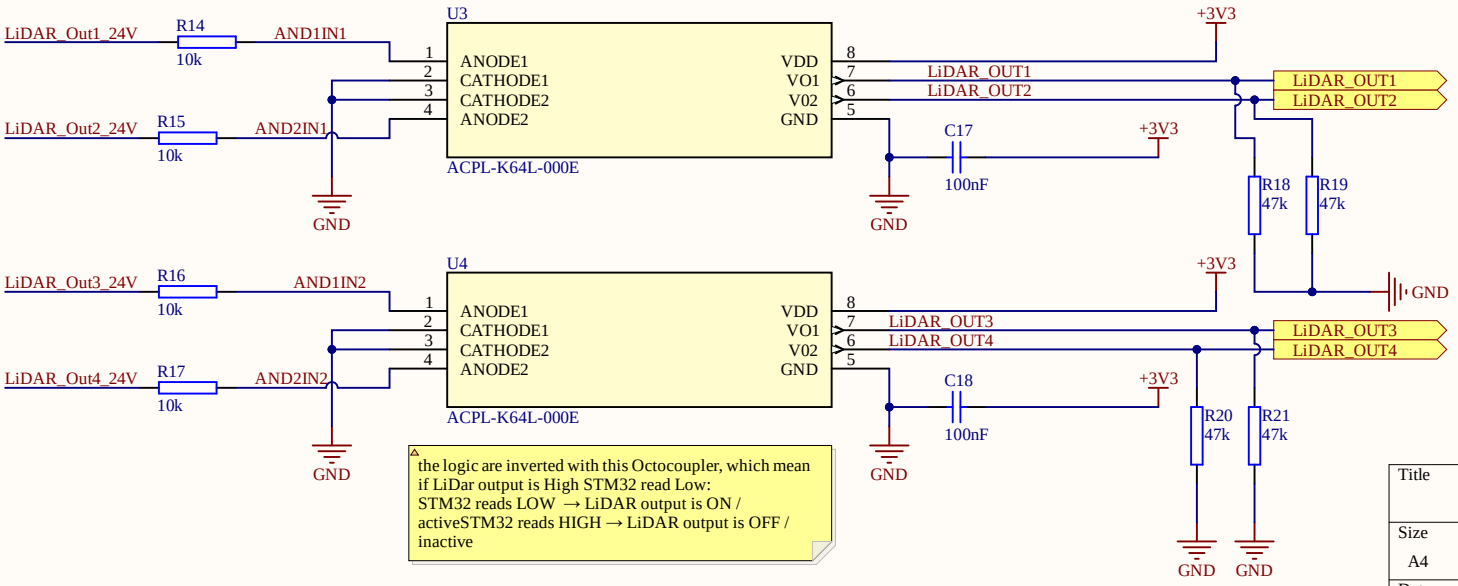


Title	Forklift LiDAR Safety Main Controller	
Size: A4	Number: 2	Revision:
Date: 10/20/2026	Sheet of 2/5	
File: C:\Users\..\MainController.SchDoc	Drawn By: Mohamed Hanan	

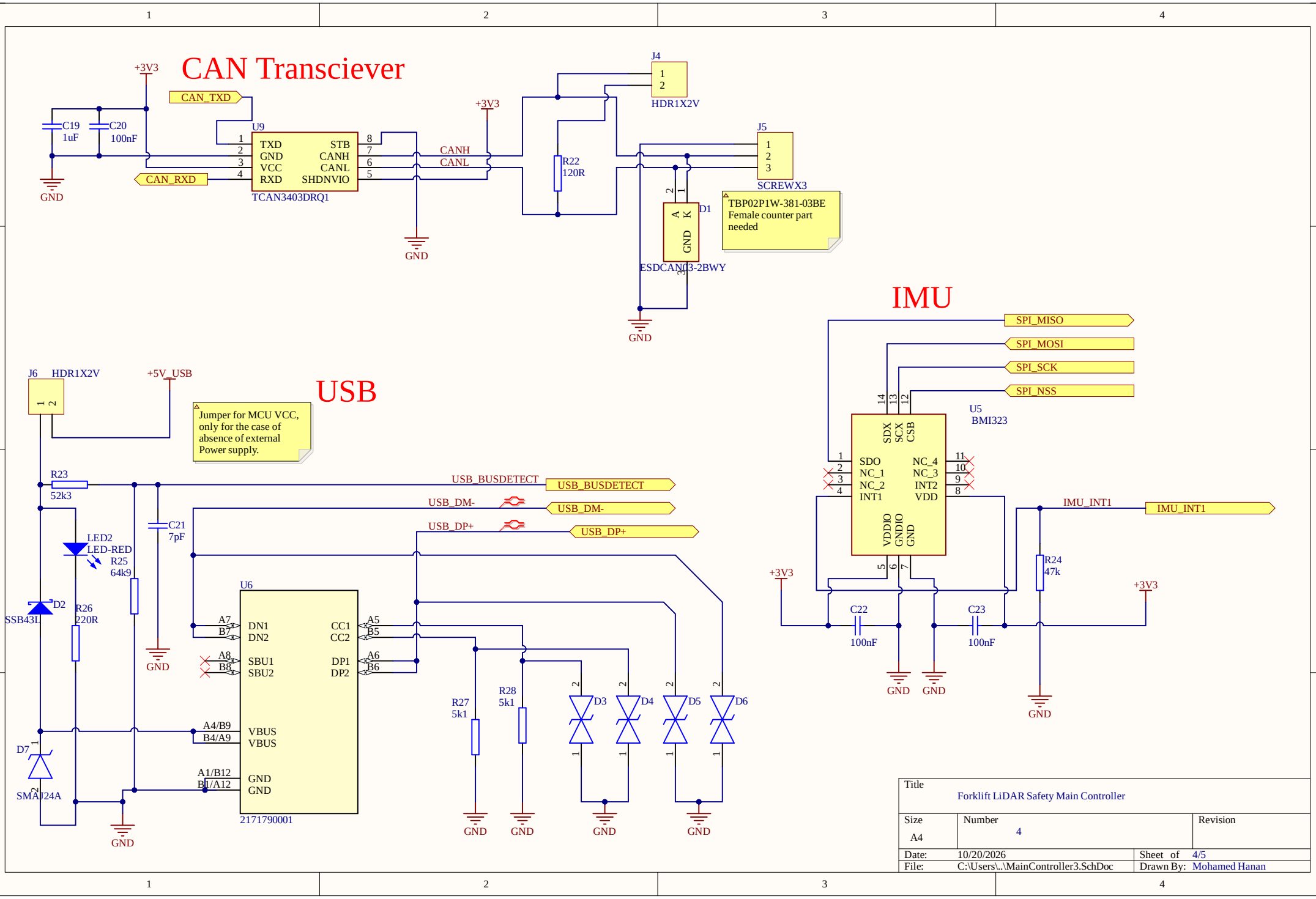
LiDAR's Field Set Slection



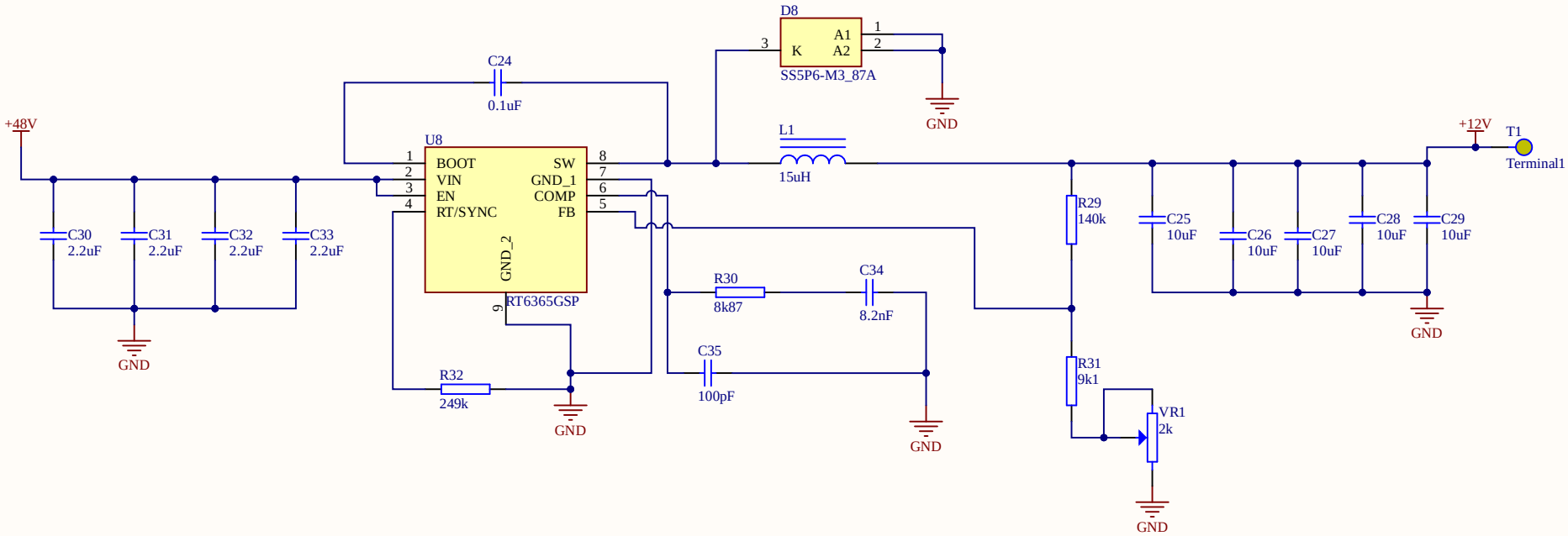
LiDAR's Logic Output



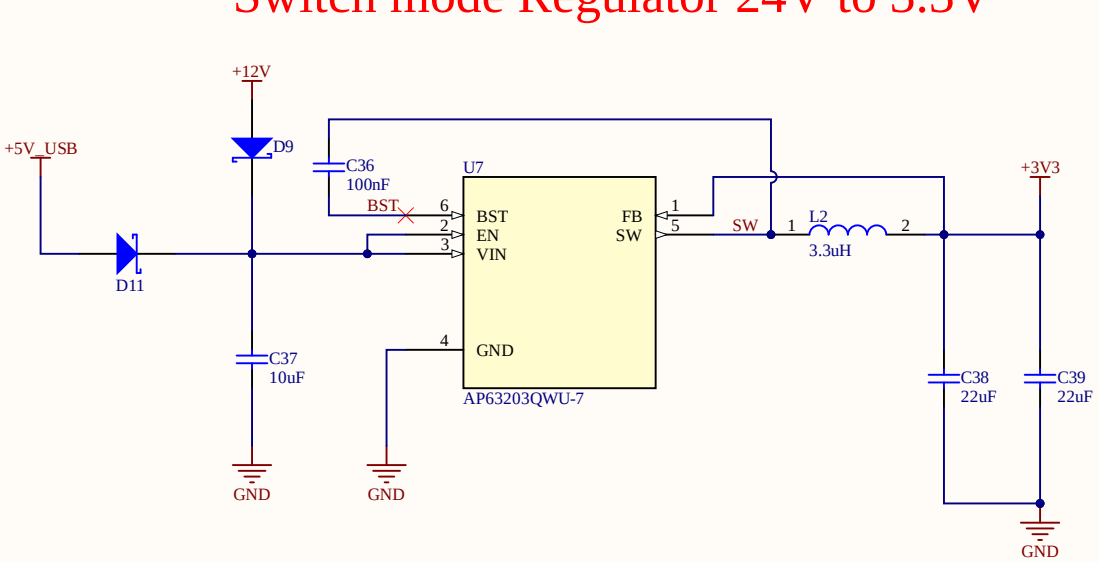
Title	Forklift LiDAR Safety Main Controller	
Size: A4	Number: 3	Revision:
Date: 10/20/2026		Sheet of 3/5
File: C:\Users\..\MainController2.SchDoc		Drawn By: Mohamed Hanan


CAN Transciever
+3V3
CAN_TXD
CAN_RXD
C19
1uF
C20
100nF
U9
TXD
GND
VCC
RXD
STB
CANH
CANL
SHDNVIO
TCAN3403DRQ1
GND
CANH
CANL
+3V3
R22
120R
J4
HDR1X2V
J5
SCREWX3
D1
ESDCAN03-2BWY
TBP02P1W-381-03BE
Female counter part
needed
GND
IMU
SPI_MISO
SPI_MOSI
SPI_SCK
SPI_NSS
U5
BMI323
SDX
SCX
CSB
SDO
NC_1
NC_2
INT1
NC_4
NC_3
INT2
VDD
VDDIO
GNDIO
GND
IMU_INT1
R24
47k
+3V3
C22
100nF
C23
100nF
GND
USB
J6
HDR1X2V
+5V_USB
Jumper for MCU VCC,
only for the case of
absence of external
Power supply.
R23
52k3
C21
7pF
USB_BUSDETECT
USB_DM-
USB_DP+
LED2
LED-RED
R25
64k9
D2
SSB43L
R26
220R
U6
DN1
DN2
SBU1
SBU2
CC1
CC2
DP1
DP2
VBUS
VBUS
GND
GND
2171790001
A7
B7
A8
B8
A5
B5
A6
B6
A4/B9
B4/A9
A1/B12
B1/A12
R27
5k1
R28
5k1
D3
D4
D5
D6
D7
SMAJ24A
GND
Title
Forklift LiDAR Safety Main Controller
Size
A4
Number
4
Revision
Date:
10/20/2026
Sheet of
4/5
File:
C:\Users\..\MainController3.SchDoc
Drawn By:
Mohamed Hanan

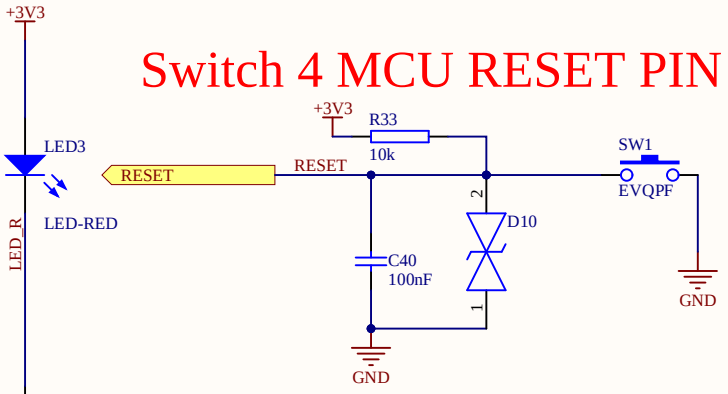
Switch mode converter form 48V to 12V



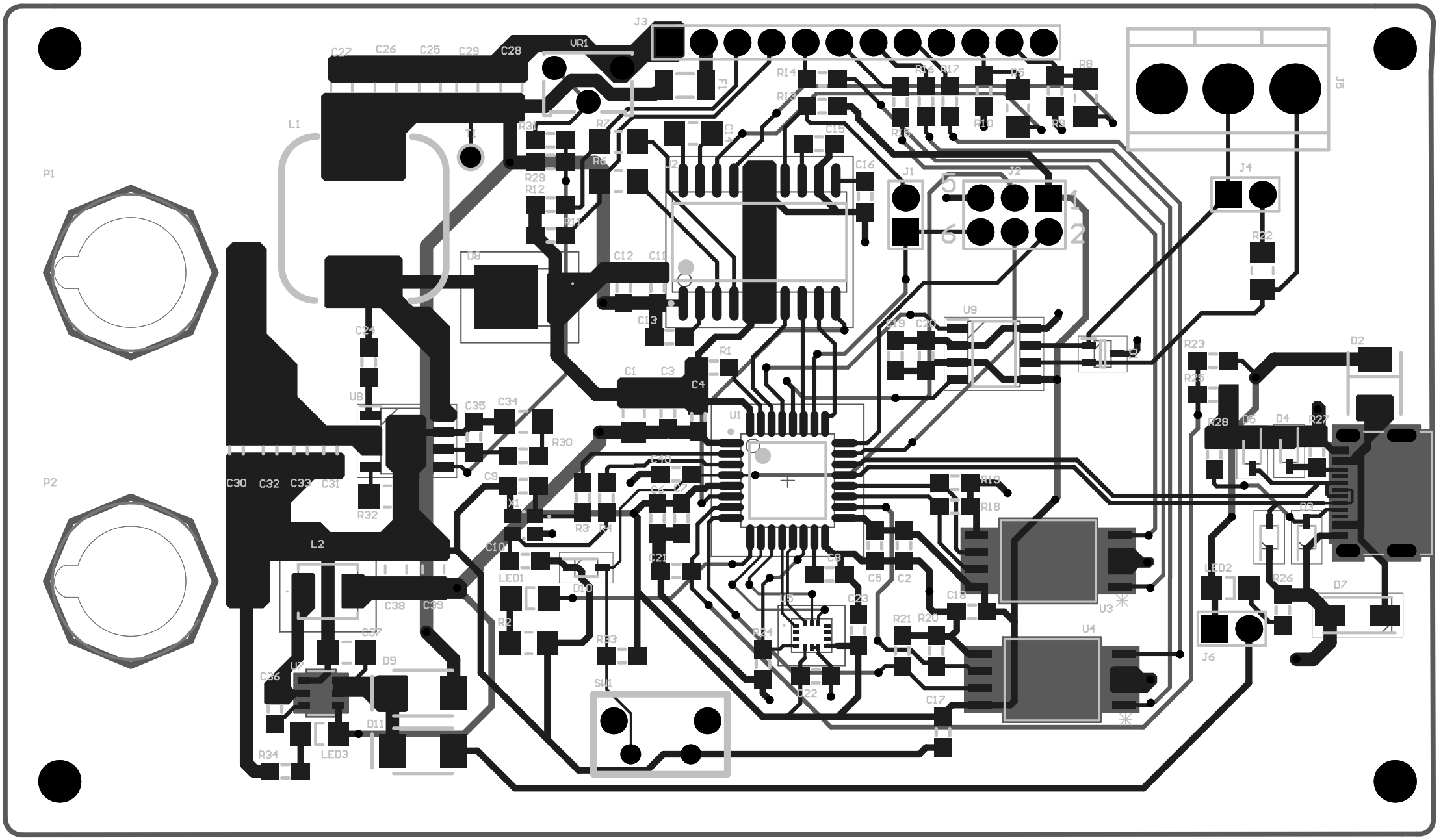
Switch mode Regulator 24V to 3.3V



Switch 4 MCU RESET PIN



Title: Forklift LiDAR Safety Main Controller		
Size: A4	Number: 5	Revision:
Date: 10/20/2026		Sheet of 5/5
File: C:\Users\..\MainController4.SchDoc		Drawn By: Mohamed Hanan


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C27
C26
C25
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C28
L1
P1
P2
J1
J2
J4
J5
J6
U1
U2
U3
U4
U8
U9
L2
LED1
LED3
SW1
D8
D9
D10
D11
C24
C30
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C1000

Comment	Description	Designator	Footprint	LibRef	Quantity
4.7uF	Non Polarized Capacitor, SMD, 1206	C1	CAP_1206	CAPN 1206	1
100nF	Non Polarized Capacitor, SMD, 0805	C2, C3, C4, C5, C7, C17, C18, C20, C22, C23, C36, C40	CAP_0805	CAPN 0805	12
1uF	Non Polarized Capacitor, SMD, 0805	C6, C12, C19	CAP_0805	CAPN 0805	3
2.2uF	Non Polarized Capacitor, SMD, 0805	C8, C30, C31, C32, C33	CAP_0805	CAPN 0805	5
7pF	Non Polarized Capacitor, SMD, 0805	C9, C10, C21	CAP_0805	CAPN 0805	3
0.1uF	Non Polarized Capacitor, SMD, 0805	C11, C24	CAP_0805	CAPN 0805	2
0.47uF	Non Polarized Capacitor, SMD, 0805	C13	CAP_0805	CAPN 0805	1
0.047uF	Non Polarized Capacitor, SMD, 1206	C14	CAP_1206	CAPN 1206	1
1nF	Non Polarized Capacitor, SMD, 0805	C15	CAP_0805	CAPN 0805	1
10nF	Non Polarized Capacitor, SMD, 0805	C16	CAP_0805	CAPN 0805	1
10uF	Non Polarized Capacitor, SMD, 1206	C25, C26, C27, C28, C29, C37	CAP_1206	CAPN 1206	6
8.2nF	Non Polarized Capacitor, SMD, 1206	C34	CAP_1206	CAPN 1206	1
100pF	Non Polarized Capacitor, SMD, 0805	C35	CAP_0805	CAPN 0805	1
22uF	Non Polarized Capacitor, SMD, 1206	C38, C39	CAP_1206	CAPN 1206	2
ESDCAN03-2BWY	TVS Diode (Uni-directional)	D1	SOT65P210X110-3N	ESDCAN03-2BWY	1
SSB43L	Schottky Diode, 4A, 30v	D2	DO-214AA(SMB)	SSB43L	1
TSD05CDYFR	TVS Diode (Bi-directional)	D3, D4, D5, D6, D10	SOD2613X100N	TSD05CDYFR	5
SMA24A	TVS Diode (Uni-directional)	D7	DIOM5127X229N	SMA24A	1
SS5P6-M3_87A	Schottky Diode	D8	SS5P6M387A	SS5P6-M3_87A	1
PMEG3050BEP	Schottky REC, 30V, 5A SMD	D9, D11	SOD-128	PMEG3050BEP	2
1206L035/16YR	Fuse, PTC Reset, 350mA Hold, 750mA Tripping, 16v	F1	MC36207	1206L035/16YR	1
HDR1X2V	Header, Plug, 2.54mm, 2 Pins, Straight	J1, J4, J5	HDR1X2_AUT	HDR1X2V	3
HDR3X2	Header, 3-Pin, Dual row, 2.54mm Pitch, Straight	J2	HDR2X3_AUT	HDR3X2	1
HDR1X12V	Header, Plug, 2.54mm, 12 Pins, Straight	J3	HDR1X12_AUT	HDR1X12V	1
SCREWX3	Terminal Block, PCB, 3Way, Pitch 5mm	J5	SCREW3_AUT	SCREWX3	1
15uH	Inductor, 15uH, 8.7A, AEC-Q200 MSS1210 Series	L1	MSS1210	MSS1210-153MED	1
3.3uH	Inductor	L2	SNR4020TA3R3M	SNR4020TA-3R3M	1
LED-RED	RED LED, Case 1206	LED1, LED2, LED3	LED1206R	SMD 1206 LED-RED	3
BS	Banana Socket, 4mm, Panel, Black, 10A	P1, P2	Banana_AUT	Banana Socket	2
10k	Resistor, SMD, 0805	R1, R9, R10, R11, R12, R13, R14, R15, R16, R17, R33	RES_0805	Resistor 0805	11
1k	Resistor, SMD, 1206	R2, R5, R6, R7, R8	RES_1206	Resistor 1206	5
0R	Resistor, SMD, 0805	R3, R4	RES_0805	Resistor 0805	2
47k	Resistor, SMD, 0805	R18, R19, R20, R21, R24	RES_0805	Resistor 0805	5
120R	Resistor, SMD, 1206	R22	RES_1206	Resistor 1206	1
52k3	Resistor, SMD, 0805	R23	RES_0805	Resistor 0805	1
64k9	Resistor, SMD, 0805	R25	RES_0805	Resistor 0805	1
220R	Resistor, SMD, 0805	R26, R34	RES_0805	Resistor 0805	2
5k1	Resistor, SMD, 0805	R27, R28	RES_0805	Resistor 0805	2
140k	Resistor, SMD, 0805	R29	RES_0805	Resistor 0805	1
8k87	Resistor, SMD, 0805	R30	RES_0805	Resistor 0805	1
9k1	Resistor, SMD, 0805	R31	RES_0805	Resistor 0805	1
249k	Resistor, SMD, 1206	R32	RES_1206	Resistor 1206	1
EVQPF	PANASONIC EVQPF203M SWITCH, TACTILE, SPST-NO, 20mA 15VDC	SW1	EVQPF203M	EVQPF	1
Terminal1	Terminal, Pad 70x70mil, Hole 32 mil	T1	PAD1_AUT	Terminal1	1
STM32C552KET6	Arm? Cortex?-M33 32-bit MCU with FPU, 144 MHz, 593 CoreMark?, up to 512Kbyte dual bank flash memory, 128-Kbyte RAM	U1	32A_M	STM32C552KET6	1
L6374FP	INDUSTRIAL QUAD LINE DRIVER	U2	20S_M	L6374FP	1
ACPL-K64L-000E	Description	U3, U4	STRETCHED_SO-8_AVA	ACPL-K64L-000E	2
BMI323	Integrated Circuit	U5	BMI323	BMI323	1
2171790001	Universal Serial Bus (USB) Shielded I/O Receptacle, Type C, Right-Angle, Top Mount, Surface Mount, Single Row, 16 Circuits, Gold (Au) Plating	U6	2171790001_MOL	2171790001	1
AP63203QWU-7	Description	U7	TSOT26_DIO	AP63203QWU-7	1
RT6365GSP	Integrated Circuit	U8	SOIC127P600X175-9N	RT6365GSP	1
TCAN3403DRQ1	Integrated Circuit	U9	SOIC127P600X175-8N - duplicate	TCAN3403DRQ1	1
2k	3362S Series, 1/4 " Square Potentiometer, 2k, 10%, 0.5 Watts, 1Turn	VR1	3362S	3362S-1-202LF	1
24MHz	Crystals, 24MHz	X1	CX2520SB	CX2520SB24000D0FLJZZ	1